

TOPOLOGICAL PRELIMINARIES FOR THE STABLE $\mathbb{A}^1$ -HOMOTOPY CATEGORY

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Goal. We want to construct the motivic homotopy category associated to a base scheme S . Recall from last quarter, let Sm/S be the category of smooth schemes of finite type over a noetherian scheme S . Let $\mathrm{Shv}(\mathrm{Sm}/S)$ be the category of sheaves with respect to the Nisnevich topology (to be recalled next week). We then localize at $\mathbb{A}^1$, so that the projection $X \times \mathbb{A}^1 \rightarrow X$ induces $F(X) \xrightarrow{\sim} F(X \times \mathbb{A}^1)$ for all $F \in \mathrm{Shv}(\mathrm{Sm}/S)$ and $X \in \mathrm{Sm}/S$.

1. ∞ -CATEGORIES CRASH COURSE

1.1. ∞ -groupoids.

Definition 1.1.1. To any topological space X we can associate a simplicial set $\mathrm{Sing}(X) : \Delta^{op} \rightarrow \mathrm{Set}$ given by $[n] \mapsto \mathrm{Hom}(|\Delta^n|, X)$.

Remark 1.1.2. $\mathrm{Sing}(X)$ satisfies the condition that for every $0 \leq k \leq n$, every morphism $f : \Lambda_k^n \rightarrow \mathrm{Sing}(X)$ from the k -th horn extends to a map $\bar{f} : \Delta^n \rightarrow \mathrm{Sing}(X)$. This is the definition of an ∞ -groupoid, or Kan complex, since every path $\gamma : [0, 1] \rightarrow X$ represented by $\gamma \in \mathrm{Sing}(X)_1$, is invertible. We then get the definition of an ∞ -category by taking lifting for only inner horns, replace the above condition for $0 < k < n$, which amounts to allowing 1-simplices (arrows) to be non-invertible.

Remark 1.1.3. The category $\mathcal{P}(\Delta)$ of simplicial sets is closed monoidal: The functor $X \times -$ admits a right adjoint $\underline{\mathrm{Hom}}(X, -)$ given by $\underline{\mathrm{Hom}}(X, Y)_n = \mathrm{Hom}(\Delta^n \times X, Y)$. And when X, Y are ∞ -categories, $\underline{\mathrm{Hom}}(X, Y)$ is an ∞ -groupoid. Since ∞ -groupoids model spaces, ∞ -categories can also be modeled as categories enriched in spaces. Additionally, for any two objects $c, d \in \mathcal{C}_0$, there is a mapping space (Kan complex) $\mathrm{Map}_{\mathcal{C}}(c, d)$ with $\mathrm{Map}_{\mathcal{C}}(c, d)_0 = \{f \in \mathcal{C}_1 : d_0 f = c, d_1 f = d\}$.

Definition 1.1.4. Let $\mathcal{C}$ be an ∞ -category. Then the homotopy category of $\mathcal{C}$, denoted $h\mathcal{C}$, is the 1-category with objects $\mathcal{C}_0$ and $\mathrm{Hom}_{h\mathcal{C}}(c, d) = \pi_0 \mathrm{Map}_{\mathcal{C}}(c, d)$.

1.2. Presheaves.

Definition 1.2.1. Let $\mathcal{C}$ be an ∞ -category. The presheaf category for $\mathcal{C}$ is the ∞ -category $\mathcal{P}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{op}, \mathcal{S})$ where $\mathcal{S}$ is the ∞ -category of Kan complexes. The corresponding Yoneda embedding $y : \mathcal{C} \rightarrow \mathcal{P}(\mathcal{C})$ is fully faithful (in the sense that the induced maps on mapping spaces are weak equivalences).

Remark 1.2.2. The usual 1-categorical business still goes through in this setting. $\mathcal{P}(\mathcal{C})$ is the free cocompletion of $\mathcal{C}$, in that for any ∞ -category $\mathcal{D}$, the restriction along y

$$\mathrm{Fun}^L(\mathcal{P}(\mathcal{C}), \mathcal{D}) \rightarrow \mathrm{Fun}(\mathcal{C}, \mathcal{D})$$

is an equivalence. In particular, the category of spaces $\mathcal{S}$ is the free cocompletion of a point. That is,

$$\mathrm{Fun}^L(\mathcal{S}, \mathcal{D}) \xrightarrow{ev_*} \mathcal{D}$$

is an equivalence (here the decoration L means colimit-preserving).

Definition 1.2.3. An ∞ -category is presentable if it is of the form $\mathrm{Ind}_\kappa(\mathcal{C}) \subseteq \mathcal{P}(\mathcal{C})$ for some small ∞ -category $\mathcal{C}$, where $\mathrm{Ind}_\kappa(\mathcal{C})$ is the subcategory of κ -filtered colimits of representables for a regular cardinal κ . Denote $\mathrm{Pr} \subseteq \mathrm{CAT}_\infty$ the subcategory of presentable ∞ -categories. Denote Pr^L the subcategory given by colimit-preserving functors. The notation comes from the fact that a presentable ∞ -category preserves colimits if and only if it is a left adjoint.

2. STABLE ∞ -CATEGORIES AND SPECTRA

2.1. (Semi-)additivity and monoids.

Definition 2.1.1. Let C be an ∞ -category with products and let $\mathbb{F}$ denote the category of finite sets. A *commutative monoid* in C is a functor

$$M : \mathbb{F}_* \rightarrow C$$

such that for every n , $M(\rho_i) : M(\langle n \rangle) \rightarrow M(\langle 1 \rangle)$ exhibits $M(\langle n \rangle) \cong (M(\langle 1 \rangle))^{\times n}$. Here $\rho_i : \langle n \rangle \rightarrow \langle 1 \rangle$ is defined by $\rho_i(j) = *$ when $i \neq j$ and $\rho_i(i) = 1$.

Then the category $\mathrm{CMon}(C)$ of commutative monoids in C is a full subcategory of $\mathrm{Fun}(\mathbb{F}_*, C)$. A monoid is grouplike if

$$M \times M \xrightarrow{(pr_1, m)} M \times M$$

is an equivalence (where $m : M \times M \rightarrow M$ is “multiplication”, induced by $\langle 2 \rangle \rightarrow \langle 1 \rangle$).

There are inclusions $\mathrm{CGrp}(C) \subseteq \mathrm{CMon}(C) \subseteq \mathrm{Fun}(\mathbb{F}_*, C)$.

Definition 2.1.2. An ∞ -category C is pointed if it has an initial object $\emptyset$ and a terminal object $*$ and the map $\emptyset \rightarrow *$ is an equivalence. Note that in this case $\mathrm{Map}_C(c, d)$ is pointed by the map factoring through zero. C is semi-additive if it has finite products and coproducts, and the map

$$\begin{pmatrix} id_X & 0 \\ 0 & id_Y \end{pmatrix} : X \times Y \rightarrow X \sqcup Y$$

is an equivalence. Write $X \oplus Y$. Note that C is semi-additive if and only if the forgetful functor $\mathrm{CMon}(C) \rightarrow C$ is an equivalence. C is additive if the shear map

$$X \oplus X \xrightarrow{(pr_1, \nabla)} X \oplus X$$

is an equivalence for all X . Similarly, this means that $\mathrm{CGrp}(C) \rightarrow C$ is an equivalence.

Remark 2.1.3. If C is semi-additive, then $\mathrm{Map}_C(c, d)$ is a commutative monoid. If C is additive, then the mapping spaces are grouplike.

Definition 2.1.4. Let C be an ∞ -category with finite products. An *associative monoid* or $\mathbb{E}_1$ -monoid in C is a simplicial object $M : \Delta^{op} \rightarrow C$ such that for every $1 \leq i \leq n$, $M(\rho_i) : M([n]) \rightarrow M([1])$ exhibits $M([n]) \simeq M([1])^{\times n}$. Denote $\mathrm{Mon}(C) = \mathrm{Mon}_{\mathbb{E}_1}(C) \subseteq \mathrm{Fun}(\Delta^{op}, C)$ the full subcategory of $\mathbb{E}_1$ -monoids, and similarly $\mathrm{Grp}_{\mathbb{E}_1}(C)$ the subcategory of associative $\mathbb{E}_1$ -groups.

Example 2.1.5. Let $X \in \mathcal{S}_*$. Then $\Omega X = \mathrm{Map}_*(S^1, X)$ is an $\mathbb{E}_1$ -group in $\mathcal{S}_*$ with the operation defined from $(S^1, *)$ being a cogroup object in $\mathcal{S}_*$ via the pinch map $S^1 \rightarrow S^1 \vee S^1$. Then $\Omega : \mathcal{S}_* \rightarrow \mathrm{Grp}(\mathcal{S})$ has a left adjoint B given by $BX = \mathrm{colim}_{\Delta^{op}} X_n$ (bar construction). This is a simplicial object where face maps are multiplication or projection and degeneracy maps include the unit.

Theorem 2.1.6. B is fully faithful, and $\mathrm{Grp}(\mathcal{S}) \simeq \mathcal{S}_*^{\geq 1}$ is the category of connected pointed spaces.

Corollary 2.1.7. The free $\mathbb{E}_1$ -group on a pointed space X is $\Omega\Sigma X$.

Proof. We have the chain of functors, with left adjoints on top:

$$\mathrm{free} : \mathcal{S}_* \rightleftarrows \mathrm{CGrp}(\mathcal{S}) : \mathrm{forget}, \quad B : \mathrm{CGrp}(\mathcal{S}) \rightleftarrows \mathcal{S}_*^{\geq 1} : \tilde{\Omega}$$

then the composition

$$\Sigma \dashv (\mathrm{forget} \circ \tilde{\Omega}) =: \Omega$$

□

Definition 2.1.8. Then define an $\mathbb{E}_n$ -monoid in $\mathcal{C}$ inductively to be $\mathrm{Mon}_{\mathbb{E}_n}(\mathcal{C}) = \mathrm{Mon}_{\mathbb{E}_{n-1}}(\mathrm{Mon}_{\mathbb{E}_1}(\mathcal{C}))$. Then $\mathrm{Grp}_{\mathbb{E}_n}(\mathcal{C}) \subseteq \mathrm{Mon}_{\mathbb{E}_n}(\mathcal{C}) \subseteq \mathrm{Fun}((\Delta^{op})^{\times n}, \mathcal{C})$ defines $\mathbb{E}_n$ -groups

Proposition 2.1.9. $\Omega^n : \mathcal{S}_*^{\geq n} \rightarrow \mathrm{Grp}_{\mathbb{E}_n}(\mathcal{S})$ is an equivalence. That is, $\mathbb{E}_n$ -groups in spaces are the same as n -connected spaces ($\pi_k = 0$ for $k < n$).

Theorem 2.1.10 (Dunn Additivity). $\mathrm{CMon}(\mathcal{C}) \simeq \lim_n \mathrm{Mon}_{\mathbb{E}_n}(\mathcal{C})$, where $\mathrm{Mon}_{\mathbb{E}_n}(\mathcal{C})$ are algebras for a little n -disc operad in $\mathcal{C}$.

Corollary 2.1.11. $\mathrm{CGrp}(\mathcal{S}) = \lim(\cdots \rightarrow \mathcal{S}_*^{\geq n} \xrightarrow{\Omega} \mathcal{S}_*^{\geq n-1} \rightarrow \cdots)$, that is, a commutative group in spaces is an infinite loop space. Equivalently, a commutative group in spaces is a sequence of pointed spaces $(X_n)_{n \in \mathbb{Z}}$, where $X_n \simeq \Omega X_{n+1}$ and X_n is n -connected.

Example 2.1.12. (1) For an abelian group A , there is an Eilenberg-Mac Lane commutative group in spaces $K(A, n)$.

(2) For $X \in \mathcal{S}_*$, the free $\mathbb{E}_1$ -group on X is $\Omega\Sigma X$. Then the free $\mathbb{E}_n$ -group is $\Omega^n \Sigma^n X$. The free commutative group on X is $\mathrm{colim}_n \Omega^n \Sigma^n X =: QX$. Note that $QS^0 = \mathrm{colim}_n \Omega^n S^n = (\bigsqcup_n B\Sigma_n)_{\mathrm{grp}}$.

2.2. The category of spectra. Spectra drops the n -connectedness to allow for infinite loop spaces whose deloopings are not n -connected for any non-negative n .

Definition 2.2.1. The ∞ -category of spectra, Sp , is defined as the inverse limit

$$\mathrm{Sp} = \lim_n (\cdots \rightarrow \mathcal{S}_* \xrightarrow{\Omega} \mathcal{S}_* \rightarrow \cdots)$$

i.e. a spectrum is a sequence of pointed spaces $X_n \in \mathcal{S}_*$ such that $X_n \simeq \Omega X_{n+1}$. The homotopy groups are then $\pi_i X = \mathrm{colim}_n \pi_{i+n} X_n$. This is well-defined since $\pi_{i+n+1} X_{n+1} \cong \pi_{i+n} \Omega X_{n+1} \cong \pi_{i+n} X_n$.

Proposition 2.2.2. $\mathrm{CGrp}(\mathcal{S}) \simeq \mathrm{Sp}^{\geq 0} \subset \mathrm{Sp}$ where $\mathrm{Sp}^{\geq 0}$ is the full subcategory of connective spectra ($\pi_i = 0$ when $i < 0$).

Remark 2.2.3. This is analogous to working $\mathbb{Z}$ -linearly in an ∞ -context. We have that $\mathrm{Mod}_{\mathbb{Z}}(\mathrm{CGrp}(\mathcal{S}))$ is equivalent to the derived ∞ -category of non-negatively graded chain complexes of abelian groups, i.e. $\mathrm{Ch}^{\geq 0}(\mathbb{Z})[\mathrm{q.iso}^{-1}]$, then $\mathrm{Mod}_{\mathbb{Z}}(\mathrm{Sp})$ is equivalent to the derived ∞ -category of unbounded chain complexes of $\mathbb{Z}$ -modules.

With all of this built up, we at last make our main definitions.

Definition 2.2.4. An ∞ -category $\mathcal{C}$ is stable if it is additive and if a square is a pushout if and only if it is a pullback. A functor between stable ∞ -categories is exact if it preserves finite limits (equivalently, colimits). We denote the ∞ -category of presentable stable ∞ -categories by $\mathrm{Pr}_{\mathrm{st}}^L \subseteq \mathrm{Pr}^L$.

Remark 2.2.5. Let $\mathcal{C}$ be an ∞ -category with finite limits and a terminal object. We can then freely form a stable ∞ -category $\mathrm{Sp}(\mathcal{C})$ from $\mathcal{C}$ by following the same construction for spaces. Let $\mathcal{C}_{*/} = \mathcal{C}_*$ be the slice, which is a pointed category. Then we define a functor $\Omega : \mathcal{C} \rightarrow \mathcal{C}$ by taking $X \mapsto \lim(* \rightarrow X \leftarrow *)$. We can then define

$$\mathrm{Sp}(\mathcal{C}) = \lim(\cdots \rightarrow \mathcal{C}_* \xrightarrow{\Omega} \mathcal{C}_* \rightarrow \cdots)$$

which comes equipped with a forgetful functor $\Omega^\infty : \mathrm{Sp}(\mathcal{C}) \rightarrow \mathcal{C}$; this has a left adjoint Σ_+^∞ . There is also a description of $\mathrm{Sp}(\mathcal{C}) \simeq \mathrm{Exc}_*(\mathcal{S}_*, \mathcal{C})$.

Proposition 2.2.6. *The ∞ -category Sp is the initial object in Pr_{st}^L . It is the free stable cocompletion on a point, in that for any stable $\mathcal{D}$ with colimits, the functor*

$$\mathrm{Fun}^L(\mathrm{Sp}, \mathcal{D}) \xrightarrow{ev_S} \mathcal{D}$$

given by evaluation at the sphere spectrum $\mathbb{S} = \Sigma^\infty S^0$.

Remark 2.2.7. We should make an important remark on the symmetric monoidal structure. $\mathcal{S}$, under $\times$, is the initial object in $\mathrm{CAlg}(\mathrm{Pr}^L)$, in the sense that for any $\mathcal{D} \in \mathrm{CAlg}(\mathrm{Pr}_{st}^L)$, $\mathrm{Fun}^{L,\otimes}(\mathcal{S}, \mathcal{D}) \simeq *$. And similarly for $(\mathcal{S}_*, \wedge) \in \mathrm{CAlg}(\mathrm{Pr}_*^L)$.

Then the symmetric monoidal structure on Sp is the unique such structure that $\Sigma^\infty : \mathcal{S}_* \rightarrow \mathrm{Sp}$ is strong monoidal and preserves colimits in each variable. Then Sp is the initial object in $\mathrm{CAlg}(\mathrm{Pr}_{st}^L)$. Write $\wedge$ for the monoidal structure on Sp .

Let's now list a few important facts about Sp .

- (1) Sp is the colimit in Pr^L of $\mathcal{S}_* \rightarrow \mathcal{S}_* \xrightarrow{\Sigma} \mathcal{S}_* \rightarrow \dots$
- (2) Sp is the initial object in $\mathrm{CAlg}(\mathrm{Pr}_{st}^L)$.
- (3) $\Sigma^\infty : \mathcal{S}_* \rightarrow \mathrm{Sp}$ is the initial way to invert $-\wedge S^1$.
- (4) A presentable ∞ -category $\mathcal{C}$ is stable if and only if $\Sigma_+^\infty \otimes \mathcal{C} : \mathcal{C} \simeq \mathcal{S} \otimes \mathcal{C} \rightarrow \mathrm{Sp} \otimes \mathcal{C}$ is an equivalence in Pr^L .

3. WHAT IS A COHOMOLOGY THEORY FOR SCHEMES?

All of this ∞ -nonsense, and yet, we are overdue in discussing cohomology theories or even what a spectrum is.

Definition 3.0.1. A spectrum E is a sequence of pointed spaces $(E_n)_{n \geq 0}$ with structure maps $\sigma_n : S^1 \wedge E_n \rightarrow E_{n+1}$ such that the adjoint $\bar{\sigma}_n : E_n \rightarrow \Omega E_{n+1}$ is an equivalence.

Brown representability tells us that every cohomology theory

$$E : \mathrm{CW}^{op} \rightarrow \mathrm{Set}$$

has a representing spectrum and conversely every spectrum E gives rise to a cohomology theory.

Example 3.0.2. (1) Singular cohomology with coefficients in A is represented by the Eilenberg-Mac Lane spectrum HA with $HA_n = K(A, n)$.

(2) Topological K -theory KU has representing spectrum

$$ku = BU \times \mathbb{Z}, \Omega BU, BU \times \mathbb{Z}, \dots$$

where BU is the classifying space for the unitary group $U = \mathrm{colim}_n U_n$.

But let's not forget the main point for the future. If we want an $\mathbb{A}^1$ -invariant homotopy and cohomology theory for schemes, we can follow the abstract (but very successful) approach above to produce the right notion of motivic spectrum. We already saw last year (recall Animesh lecture 9), that the most obvious way of defining an $\mathbb{A}^1$ -homotopy of schemes has its issues. We will do this by defining motivic spaces, pass to pointed objects, and stabilize. We'll then see that much of the cohomology theory is captured by the *six functor formalism*.

4. MINI APPENDIX: MODEL CATEGORIES

We should also say a word about the model categorical structure on motivic homotopy theory. A key computational and conceptual tool for classical homotopy theory is understood from fibrations and cofibrations. This is formalized as a model structure on the category of spaces. This is the data in a category $\mathcal{C}$ with finite limits and colimits of classes of morphisms WE, Cof, Fib satisfying certain axioms. One of the axioms is that every arrow in $\mathcal{C}$ can be both factored as an arrow in $WE \cap Cof$ followed by one in Fib or as

an arrow in Cof followed by one in $WE \cap Fib$. The interval in homotopy theory plays a formal role here: The interval $I = [0, 1]$ is unique up to homotopy such that the fold map $* \sqcup * \xrightarrow{fold} *$ factors as

$$* \sqcup * \hookrightarrow I \xrightarrow{\sim} *$$

We can factor $fold : X \sqcup X \rightarrow X$ for any object X gives a notion of left homotopy. In algebraic geometry, we want $\mathbb{A}^1$ to play this role.

Dually however, for any $X \in \mathcal{C}$ we can factor the diagonal $\Delta : X \rightarrow X \times X$ as

$$X \xrightarrow{\sim} PX \xrightarrow{(d_0, d_1)} X \times X$$

Then pulling back allows us to form loop space objects (which define right homotopies)

$$\begin{array}{ccc} \Omega X & \longrightarrow & PX \\ \downarrow & \lrcorner & \downarrow (d_0, d_1) \\ X & \xrightarrow{\Delta} & X \times X \end{array}$$

If we want to understand this in the algebraic setting, we follow from topology. $\Omega X \in CW$ is represented by S^1 , in that $\Omega X = Map_*(S^1, X)$, and $PX = Map(I, X)$. It isn't always the case that the path space on X is the exponential of the interval object, but it does provide some guidance on what should replace in algebraic geometry, the very important objects in the classical setting, $S^1, I, *, \Omega, \Sigma, \dots$